

Assignment No: 10

Title: Modules and Packages

Objective: To understand modules and packages in Python

The assignment covers:

Course Outcome-CO5

Bloom's Cognitive Domain-Creating

Basic Theory:

A **module** allows you to logically organize your Python code. Grouping related code into a module makes the code easier to understand and use. A module is a Python object with arbitrarily named attributes that you can bind and reference.

Simply, a module is a file consisting of Python code. A module can define functions, classes and variables. A module can also include runnable code.

The import has the following syntax –

```
import module1[, module2[,... moduleN]
```

The from...import * Statement

It is also possible to import all names from a module into the current namespace by using the following import statement –

```
from modname import *
```

Packages in Python

A package is a hierarchical file directory structure that defines a single Python application environment that consists of modules and subpackages and sub-subpackages, and so on.

Problem Statement:

1. Write a program which returns the year and name of weekday

```
import datetime
```

```
x = datetime.datetime.now()
```

```
print(x.year)
```

```
print(x.strftime("%A"))
```

Output:

2019

Tuesday

2. Write a module for calculator. Perform all the basic mathematical operations using the created module.

```
mod.py
```

```
def add(x,y):
```

```
    return x+y
```

```
def sub(x,y):
```

```
    return x-y
```

```
def mul(x,y):
```

```
    return x*y
```

```
def div(x,y):
```

```
    return x/y,x%y
```

```
import mod
```

```
x=int(input("Enter a Number:"))
```

```
y=int(input("Enter a Number:"))
```

```
print(mod.add(x,y))
```

```
print(mod.sub(x,y))
```

```
print(mod.mul(x,y))
```

```
print(mod.div(x,y))
```

Output:

Enter a Number:5

Enter a Number:2

7

3

10

(2, 1)

3. Print all the modules, variables and functions that are defined under module random.

```
>>>dir(random)
```

```
['BPF', 'LOG4', 'NV_MAGICCONST', 'RECIP_BPF', 'Random', 'SG_MAGICCONST',  
'SystemRandom', 'TWOPI', 'WichmannHill', '_BuiltinMethodType', '_MethodType', '__all__',  
'__builtins__', '__doc__', '__file__', '__name__', '__package__', '_acos', '_ceil', '_cos', '_e',  
'_exp', '_hashlib', '_hexlify', '_inst', '_log', '_pi', '_random', '_sin', '_sqrt', '_test',  
'_test_generator', '_urandom', '_warn', 'betavariate', 'choice', 'division', 'expovariate',  
'gammavariate', 'gauss', 'getrandbits', 'getstate', 'jumpahead', 'lognormvariate',  
'normalvariate', 'paretovariate', 'randint', 'random', 'randrange', 'sample', 'seed', 'setstate',  
'shuffle', 'triangular', 'uniform', 'vonmisesvariate', 'weibullvariate']
```

4. Import datetime module and print current date and time.

```
import datetime  
x = datetime.datetime.now()  
print(x)
```

Output:

2019-08-20 17:03:35.329094

5. Import Calendar module and print the calendar of any year.

```
import calendar  
  
print ("The calender of year 2012 is : ")  
  
print (calendar.calendar(2012,2,1,6))
```

Output:

2012

January

February

March

Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su

1 1 2 3 4 5 1 2 3 4
2 3 4 5 6 7 8 6 7 8 9 10 11 12 5 6 7 8 9 10 11
9 10 11 12 13 14 15 13 14 15 16 17 18 19 12 13 14 15 16 17 18
16 17 18 19 20 21 22 20 21 22 23 24 25 26 19 20 21 22 23 24 25
23 24 25 26 27 28 29 27 28 29 26 27 28 29 30 31
30 31

April May June

Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su

1 1 2 3 4 5 6 1 2 3
2 3 4 5 6 7 8 7 8 9 10 11 12 13 4 5 6 7 8 9 10
9 10 11 12 13 14 15 14 15 16 17 18 19 20 11 12 13 14 15 16 17
16 17 18 19 20 21 22 21 22 23 24 25 26 27 18 19 20 21 22 23 24
23 24 25 26 27 28 29 28 29 30 31 25 26 27 28 29 30
30

July August September

Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su

1 1 2 3 4 5 1 2
2 3 4 5 6 7 8 6 7 8 9 10 11 12 3 4 5 6 7 8 9
9 10 11 12 13 14 15 13 14 15 16 17 18 19 10 11 12 13 14 15 16
16 17 18 19 20 21 22 20 21 22 23 24 25 26 17 18 19 20 21 22 23
23 24 25 26 27 28 29 27 28 29 30 31 24 25 26 27 28 29 30
30 31

October November December

Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa Su

1 2 3 4 5 6 7 1 2 3 4 1 2

8 9 10 11 12 13 14	5 6 7 8 9 10 11	3 4 5 6 7 8 9
15 16 17 18 19 20 21	12 13 14 15 16 17 18	10 11 12 13 14 15 16
22 23 24 25 26 27 28	19 20 21 22 23 24 25	17 18 19 20 21 22 23
29 30 31	26 27 28 29 30	24 25 26 27 28 29 30
31		

6. Check whether a given year is leap year or not. Provide a range of years and display how many leap years are there.

```
import calendar

y=int(input("Enter a Year:"))

st= int(input("Enter Start Year:"))

end= int(input("Enter End Year:"))

if (calendar.isleap(y)):

    print ("The year is leap")

else : print ("The year is not leap")

print ("The leap days between : %d and %d is"%(st,end),end="")

print (calendar.leapdays(st,end))
```

Output:

Enter a Year: 2008

Enter Start Year: 1950

Enter End Year: 2000

The year is leap

The leap days between 1950 and 2000 are : 12

7. Print a month by importing calendar module.

```
import calendar

# using month() to display month of specific year

print ("The month 5th of 2016 is :")

print (calendar.month(2016,5,2,1))
```

Output:

The month 5th of 2016 is :

May 2016

Mo Tu We Th Fr Sa Su

1

2 3 4 5 6 7 8

9 10 11 12 13 14 15

16 17 18 19 20 21 22

23 24 25 26 27 28 29

30 31

8. Write a program to print a given datetime.

```
import datetime
```

```
x = datetime.datetime(2020, 5, 17)
```

```
print(x)
```

Output:

2020-05-17 00:00:00

9. Display the name of the month.

```
import datetime
```

```
x = datetime.datetime(2018, 6, 1)
```

```
print(x.strftime("%B"))
```

Output:

August

10. Print the month names of calendar.

```
import calendar
```

```
for name in calendar.month_name:
```

```
    print(name)
```

Output:

January

February

March

April

May

June

July

August

September

October

November

December

11. Print the name of days in a week.

```
import calendar
```

```
for day in calendar.day_name:
```

```
    print(day)
```

Output:

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

12. Write a program which will print the month name and first Monday in a given year.

```
import calendar
```

```

yy=int(input("Enter a Year: "))
for month in range (1,13):
    #it retrieves a list of weeks that represent the month
    mycal=calendar.monthcalendar(yy,month)
    week1=mycal[0]
    week2=mycal[1]
    if week1[calendar.Monday]!=0:
        auditday=week1[calendar.Monday]
    else:
        auditday=week2[calendar.Monday]
    print("%10s %2d"%(calendar.month_name[month],auditday))

```

Output:

Enter a Year: 2025

January 6

February 3

March 3

April 7

May 5

June 2

July 7

August 4

September 1

October 6

November 3

December 1

13. Print the current date and time in the following format.

year: 2018

month: 12

day: 24

time: 04:59:31

date and time: 12/24/2018, 04:59:31

```
from datetime import datetime
```

```
now = datetime.now() # current date and time
```

```
year = now.strftime("%Y")
```

```
print("year:", year)
```

```
month = now.strftime("%m")
```

```
print("month:", month)
```

```
day = now.strftime("%d")
```

```
print("day:", day)
```

```
time = now.strftime("%H:%M:%S")
```

```
print("time:", time)
```

```
date_time = now.strftime("%m/%d/%Y, %H:%M:%S")
```

```
print("date and time:",date_time)
```

Output:

year: 2018

month: 12

day: 24

time: 04:59:31

date and time: 12/24/2018, 04:59:31

14. Write a Python program importing random which outputs the frequency of occurrence of each faces for rolling a dice 6000 times.

```
import random
```

```
frequency1 = 0
```

```
frequency2 = 0
```

```
frequency3 = 0
frequency4 = 0
frequency5 = 0
frequency6 = 0
for roll in range( 1, 6001 ): # 6000 die rolls
    face = random.randrange( 1, 7 )
    if face == 1: # frequency counted
        frequency1 += 1
    elif face == 2:
        frequency2 += 1
    elif face == 3:
        frequency3 += 1
    elif face == 4:
        frequency4 += 1
    elif face == 5:
        frequency5 += 1
    elif face == 6:
        frequency6 += 1
    else: # simple error handling
        print "should never get here!"

print "Face %13s" % "Frequency"
print " 1 %13d" % frequency1
print " 2 %13d" % frequency2
print " 3 %13d" % frequency3
print " 4 %13d" % frequency4
print " 5 %13d" % frequency5
print " 6 %13d" % frequency6
```

Output

Face	Frequency
1	946
2	1003
3	1035
4	1012
5	987
6	1017

15. Create a module which contains the functions as stated below.

a) $F(x,y)=F(x-y,y)+1$, if $y \leq x$

b) $F(n,r)=F(n-1,r)+F(n-1,r-1)$

c) $F(n)=F(n/2)+1$ if $n>1$

d) $F(M,N)=1$ if $M=0$, or $M \geq N \geq 1$, and $F(M,N)=F(M-1,N)+F(M-1,N-1)$, otherwise.

e) $B(m,x)=m!/(x!(m-x)!)$ where $m>x$,

$B(0,0)=B(m,0)=1$ and $B(m,x)=B(m,x-1)*[(m-x+1)/x]$

module_fun.py

```
def f1(x,y):
```

```
    if y<=x:
```

```
        return f1(x-y,y)+1
```

```
    return 1
```

```
def f2(n,r):
```

```
    if n>0 and r>0:
```

```
        return (f2(n-1,r)+f2(n-1,r-1))
```

```
    else:
```

```
        return n+r
```

```
def f3(n):
```

```
    if n>1:
```

```

        return f3(n/2)+1

    return 0

def f4(m,n):

    if m==0 or (m>=n and n>=1):

        return 1

    else:

        return f4(m-1,n)+f4(m-1,n-1)

def fact(n):

    p=1

    for i in range(1,n+1):

        p*=i

    return p

def f5(m,x):

    if x==0:

        return 1

    if m>x:

        return (fact(m)/(fact(x)*fact(m-x)))

    else:

        return (f5(m,x-1)*((m-x+1)/x))

```

main.py

```
import module_fun as mf
```

```

while(1):

    print("Press i to access function number i(eg:-1 for fun1)\nPress 6 to exit")

    c=int(input())

    if c==1:

        print("Enter x and y:")

        x,y=input().split()

```

```
x=int(x)
y=int(y)
print("Result=",mf.f1(x,y))
elif c==2:
    print("Enter n and r:")
    n,r=input().split()
    n=int(n)
    r=int(r)
    print("Result=",mf.f2(n,r))
elif c==3:
    n=int(input("Enter n:"))
    print("Result=",mf.f3(n))
elif c==4:
    print("Enter m and n")
    m,n=input().split()
    m=int(m)
    n=int(n)
    print("Result=",mf.f4(m,n))
elif c==5:
    print("Enter m and x")
    m,x=input().split()
    m=int(m)
    x=int(n)
    print("Result=",mf.f5(m,x))
elif c==6:
    print("Thanks")
    break
else:
    print("Wrong input")
```

Output

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

1

Enter x and y:

5

7

('Result=', 1)

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

2

Enter n and r:

6

4

('Result=', 78)

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

3

Enter n:7

('Result=', 2)

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

4

Enter m and n

4

8

('Result=', 16)

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

5

Enter m and x

4

6

('Result=', 0)

Press i to access function number i(eg:-1 for fun1)

Press 6 to exit

6

Thanks

16. Create a module which contains the functions printing several patterns. Each function will take input the number of rows and/or columns as required.

*	**	4 5 6
**	***	7 8 9 10
***	****	11 12 13 14
****	*****	15

	1
*	1 2
**	1 2 3
***	1 2 3 4
****	1 2 3 4 5

	1
*	2 3

pattern.py

```
def pypart(n):
```

```
    # outer loop to handle number of rows
```

```
    # n in this case
```

```
    for i in range(0, n):
```

```
        # inner loop to handle number of columns
```

```
        # values changing acc. to outer loop
```

```
        for j in range(0, i+1):
```

```
            # printing stars
```

```
            print("* ",end="")
```

```
        # ending line after each row
```

```
        print("\r")
```

```
def pypart2(n):
```

```
    # number of spaces
```

```
    k = 2*n - 2
```

```
    # outer loop to handle number of rows
```

```
    for i in range(0, n):
```

```
        # inner loop to handle number spaces
```

```
        # values changing acc. to requirement
```

```
        for j in range(0, k):
```

```
            print(end=" ")
```

```
        # decrementing k after each loop
```

```
        k = k - 2
```

```
    # inner loop to handle number of columns
```

```
    # values changing acc. to outer loop
```

```
for j in range(0, i+1):  
    # printing stars  
    print("* ", end="")  
  
    # ending line after each row  
    print("\n")
```

```
def triangle(n):  
    # number of spaces  
    k = 2*n - 2  
  
    # outer loop to handle number of rows  
    for i in range(0, n):  
        # inner loop to handle number spaces  
        # values changing acc. to requirement  
        for j in range(0, k):  
            print(end=" ")  
  
        # decrementing k after each loop  
        k = k - 1  
  
        # inner loop to handle number of columns  
        # values changing acc. to outer loop  
        for j in range(0, i+1):  
            # printing stars  
            print("* ", end="")  
  
        # ending line after each row  
        print("\n")
```

```
def numpat(n):  
    # initialising starting number  
    num = 1
```

```
# outer loop to handle number of rows
for i in range(0, n):
    # re assigning num
    num = 1
    # inner loop to handle number of columns
    # values changing acc. to outer loop
    for j in range(0, i+1):
        # printing number
        print(num, end=" ")
        # incrementing number at each column
        num = num + 1
    # ending line after each row
    print("\r")
```

```
def contnum(n):
    # initializing starting number
    num = 1
    # outer loop to handle number of rows
    for i in range(0, n):
        # not re assigning num
        # num = 1
        # inner loop to handle number of columns
        # values changing acc. to outer loop
        for j in range(0, i+1):
            # printing number
            print(num, end=" ")
            # incrementing number at each column
            num = num + 1
```

ending line after each row

print("\r")

main.py

pattern.py part(5)

pattern.py part2(5)

pattern.triangle(5)

pattern.numpat(5)

pattern.contnum(5)

Output:

*

* *

* * *

* * * *

* * * * *

*

* *

* * *

* * * *

* * * * *

*

* *

* * *

* * * *

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

17. Create a package which contains five modules fish, birds, amphibians, mammals, and reptiles. Each module contains two functions example and characteristics.

amphibians.py

```
def examples():
```

```
    print("Here are some examples of amphibians:")
```

```
    eggs=['Frog',"Salamander","Toads","Newts","caecilians"]
```

```
    for f in eggs:
```

```
        print(f)
```

```
def chars():
```

```
    ch=["Cold blooded","Lays eggs","Moist scaleless skin"]
```

```
    print("Characteristics of amphibians:")
```

```
    for c in ch:
```

```
        print(c)
```

birds.py

```
def examples():
```

```
    print("Here are some examples of birds:")
```

```
    eggs=['Parrot','Pigeon','Crow','Owl','Sparrow']
```

```
    for f in eggs:
```

```
        print(f)
```

```
def chars():
```

```
    ch=["Have wings","Lays eggs","Warm blooded","Have beaks and no teeth"]
```

```
    print("Characteristics of birds:")
```

```
    for c in ch:
```

```
        print(c)
```

fish.py

```
def examples():
```

```
    print("Here are some examples of fish:")
```

```
    eggs=['Goldfish','Tuna','Guppy','Piranha','Swordfish']
```

```
    for f in eggs:
```

```
        print(f)
```

```
def chars():
```

```
    ch=["Lives Under Water","Breathes through gills","Swims using fins and Tail"]
```

```
    print("Characteristics of fish:")
```

```
    for c in ch:
```

```
        print(c)
```

mammals.py

```
def examples():
```

```
    print("Here are some examples of mammals:")
```

```

    egs=["Human","Cat","Tiger","Dog","Elephant"]

    for f in egs:

        print(f)
def chars():

    ch=["Warm blooded","Gives birth to babies","Have hair and fur","Four chambered
heart"]

    print("Characteristics of mammals:")

    for c in ch:

        print(c)

```

reptiles.py

```

def examples():

    print("Here are some examples of reptiles:")

    egs=['Turtle',"Lizard","Crocodiles","Chameleon","Snakes"]

    for f in egs:

        print(f)
def chars():

    ch=["Have scales","Vertebrates","Breathe through lungs","Cold blooded"]

    print("Characteristics of reptiles:")

    for c in ch:

        print(c)

```

Creatures.py

```

import fish,birds,amphibians,mammals,reptiles

def __init__(self):

    print("Find your information about creatures:examples and characteristics")

```

main.py

```
#Creatures package includes various modules

import Creatures

print("Find your information about creatures:examples and characteristics")

while(1):

    print("Press\n1 for fish\n2 for birds\n3 for amphibians\n4 for mammals\n5 for reptiles\n6 to exit:")

    c=int(input())

    if c==1:

        Creatures.fish.examples()

        Creatures.fish.chars()

    elif c==2:

        Creatures.birds.examples()

        Creatures.birds.chars()

    elif c==3:

        Creatures.amphibians.examples()

        Creatures.amphibians.chars()

    elif c==4:

        Creatures.mammals.examples()

        Creatures.mammals.chars()

    elif c==5:

        Creatures.reptiles.examples()

        Creatures.reptiles.chars()

    elif c==6:

        print("Thank You!!!")

        break

    else:

        print("Wrong input")
```

#for each module class concept has been introduced purposefully.

#to be noted same process can be implemented without introducing separate class

Output

Find your information about creatures:examples and characteristics

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

1

Here are some examples of fish:

Goldfish

Tuna

Guppy

Piranha

Swordfish

Characteristics of fish:

Lives Under Water

Breathes through gills

Swims using fins and Tail

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

2

Here are some examples of birds:

Parrot

Pigeon

Crow

Owl

Sparrow

Characteristics of birds:

Have wings

Lays eggs

Warm blooded

Have beaks and no teeth

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

3

Here are some examples of amphibians:

Frog

Salamander

Toads

Newts

caecilians

Characteristics of amphibians:

Cold blooded

Lays eggs

Moist scaleless skin

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

4

Here are some examples of mammals:

Human

Cat

Tiger

Dog

Elephant

Characteristics of mammals:

Warm blooded

Gives birth to babies

Have hair and fur

Four chambered heart

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

5

Here are some examples of reptiles:

Turtle

Lizard

Crocodiles

Chameleon

Snakes

Characteristics of reptiles:

Have scales

Vertebrates

Breathe through lungs

Cold blooded

Press

1 for fish

2 for birds

3 for amphibians

4 for mammals

5 for reptiles

6 to exit:

6

Thank You!!!

Discussions:

This assignment covers the basics of files, modules, and packages in Python which covers PO1, PO2, PO3, PO12, and PSO1, PSO2, PSO3.

Questionnaires:

1. What is pip?

PIP is a package manager for Python packages, or modules if you like. If you have Python version 3.4 or later, PIP is included by default.

2. How do you check whether it is installed or not?

Typing pip in the command prompt.

3. How do you check which packages of Python are installed?

pip list

